

Solutions to Problem Set 9

1. Covariance and correlation

- (a) Let $X \sim \mathcal{U}([0, a])$ for some $a > 0$. Determine the covariance $\text{Cov}[X, X^2]$ and the correlation coefficient $\rho(X, X^2)$.
- (b) Consider rolling a fair die twice, and let X_1 and X_2 denote the outcome of the die on the first and second round, respectively. We define $S = X_1 + X_2$ (sum) and $M = \max\{X_1, X_2\}$ (highest outcome).
- (i) Write down the joint probability mass function $p_{S,M}$ in a table.
- (ii) Calculate $\text{Cov}[S, M]$ and $\rho(S, M)$.
- (c) Let $X, Y \sim \text{Ber}(p)$ independent random variables, where $p \in (0, 1)$. Show that

$$\text{Cov}[X + Y, X - Y] = 0,$$

but $X + Y$ and $X - Y$ are not independent.

- (a) We have, for $n \in \mathbb{N}$,

$$\mathbf{E}[X^n] = \frac{1}{a} \int_0^a x^n dx = \frac{a^n}{n+1}.$$

$$\mathbf{E}[X] = a/2, \quad \text{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \frac{a^2}{3} - \frac{a^2}{4} = \frac{a^2}{12}.$$

Moreover,

$$\text{Var}[X^2] = \mathbf{E}[X^4] - \mathbf{E}[X^2]^2 = \frac{a^4}{5} - \frac{a^4}{9} = \frac{4}{45}a^4.$$

It follows that

$$\text{Cov}[X, X^2] = \mathbf{E}[X^3] - \mathbf{E}[X]\mathbf{E}[X^2] = \frac{a^3}{4} - \frac{a}{2} \cdot \frac{a^2}{3} = \frac{a^3}{12},$$

and

$$\rho(X, X^2) = \frac{\text{Cov}[X, X^2]}{\sqrt{\text{Var}[X]} \cdot \sqrt{\text{Var}[X^2]}} = \frac{a^3/12}{\sqrt{a^2/12} \cdot \sqrt{4a^4/45}} = \frac{\sqrt{15}}{4}.$$

- (b) (i) The following table shows the joint probability mass function $p_{S,M}$

$s_i \setminus m_j$	1	2	3	4	5	6	$p_S(s_i)$
2	1/36	0	0	0	0	0	1/36
3	0	1/18	0	0	0	0	2/36
4	0	1/36	1/18	0	0	0	3/36
5	0	0	1/18	1/18	0	0	4/36
6	0	0	1/36	1/18	1/18	0	5/36
7	0	0	0	1/18	1/18	1/18	6/36
8	0	0	0	1/36	1/18	1/18	5/36
9	0	0	0	0	1/18	1/18	4/36
10	0	0	0	0	1/36	1/18	3/36
11	0	0	0	0	0	1/18	2/36
12	0	0	0	0	0	1/36	1/36
$p_M(m_j)$	1/36	3/36	5/36	7/36	9/36	11/36	1

(ii) We now calculate

$$\mathbf{E}[S] = \sum_{s=2}^{12} s \cdot p_S(s) = 7,$$

which can also be obtained by setting $S = X_1 + X_2$ with $X_1, X_2 \sim \mathcal{U}(\{1, \dots, 6\})$ i.i.d. and $\mathbf{E}[S] = 2\mathbf{E}[X_1] = 2 \cdot 3.5 = 7$.

$$\mathbf{E}[M] = \sum_{m=1}^6 m \cdot p_M(m) = \frac{161}{36}.$$

$$\mathbf{E}[MS] = \sum_{m=1}^6 \sum_{s=2}^{12} ms \mathbf{P}[M = m, S = s] = \frac{308}{9}.$$

Moreover (again with $S = X_1 + X_2$ with $X_1, X_2 \sim \mathcal{U}(\{1, \dots, 6\})$ i.i.d.),

$$\text{Var}[S] = \text{Var}[X_1 + X_2] = 2\text{Var}[X_1] = \frac{105}{18}.$$

$$\text{Var}[M] = \sum_{m=1}^6 m^2 \mathbf{P}[M = m] - (\mathbf{E}[M])^2 = \frac{2555}{1296}.$$

We can calculate

$$\text{Cov}[S, M] = \mathbf{E}[SM] - \mathbf{E}[S]\mathbf{E}[M] = \frac{308}{9} - 7 \cdot \frac{161}{36} = \frac{35}{12}.$$

$$\rho(S, M) = \frac{\frac{35}{12}}{\sqrt{\frac{105}{18}} \sqrt{\frac{2555}{1296}}} \approx 0.8601.$$

(c) We use the bilinearity of the covariance:

$$\begin{aligned} \text{Cov}[X + Y, X - Y] &= \text{Cov}[X, X] - \underbrace{\text{Cov}[X, Y]}_{=0} + \underbrace{\text{Cov}[Y, X]}_{=0} - \text{Cov}[Y, Y] \\ &= p(1 - p) - p(1 - p) = 0. \end{aligned}$$

In the second line we used that $\text{Cov}[X, X] = \text{Var}[X] = p(1 - p) = \text{Var}[Y]$, since $X \stackrel{d}{=} Y \sim \text{Ber}(p)$. However, $X + Y$ and $X - Y$ are not independent. Indeed:

$$\mathbf{P}[X + Y = 2, X - Y = 1] = 0,$$

whereas

$$\begin{aligned} \mathbf{P}[X + Y = 2] &= \mathbf{P}[\{X = 1\} \cap \{Y = 1\}] = p^2, \\ \mathbf{P}[X - Y = 1] &= \mathbf{P}[\{X = 1\} \cap \{Y = 0\}] = p(1 - p). \end{aligned}$$

In particular, we have

$$\mathbf{P}[X + Y = 2, X - Y = 1] \neq \mathbf{P}[X + Y = 2]\mathbf{P}[X - Y = 1],$$

so $X + Y$ and $X - Y$ are not independent.

2. (*) Decay of covariances in an autoregressive process

[4 Points]

Let $\sigma, \alpha \in \mathbb{R}$, $|\alpha| < 1$. We consider random variables X_0 and $(Z_n)_{n \in \mathbb{N}}$, independent and assume that $Z_n \sim \mathcal{N}(0, 1)$. We define a so-called *autoregressive process* (also called AR(1)-process) inductively by

$$X_n = \alpha X_{n-1} + \sigma Z_n, \quad n \in \mathbb{N}.$$

Prove the following claims.

- (a) If $X_{n-1} \sim \mathcal{N}(m, v)$, then $X_n \sim \mathcal{N}(\alpha m, \alpha^2 v + \sigma^2)$.
(b) The distribution $\mathcal{N}(0, \sigma^2(1 - \alpha^2)^{-1})$ is *stationary*, meaning that

$$X_0 \sim \mu \quad \Rightarrow \quad X_n \sim \mu \text{ for all } n \in \mathbb{N}.$$

- (c) Assuming that $X_0 \sim \mu$, show that

$$\text{Cov}[X_n, X_{n-k}] = \alpha^k \frac{\sigma^2}{1 - \alpha^2}, \quad 0 \leq k \leq n.$$

- (a) From (5.18) in the notes:

$$\alpha X_{n-1} \sim \mathcal{N}(\alpha m, \alpha^2 v), \quad \sigma Z_n \sim \mathcal{N}(0, \sigma^2).$$

Moreover, the random variables αX_{n-1} and σZ_n are still independent. Therefore, by Problem 2 on Problem set 8:

$$X_n = \alpha X_{n-1} + \sigma Z_n \sim \mathcal{N}(\alpha m, \alpha^2 v + \sigma^2).$$

- (b) Inserting $m = 0, v = \sigma^2/(1 - \alpha^2)$ in (a), we have

$$X_{n-1} \sim \mathcal{N}(0, \sigma^2/(1 - \alpha^2)) \Rightarrow X_n \sim \mathcal{N}(0, \alpha^2 \sigma^2/(1 - \alpha^2) + \sigma^2) = \mathcal{N}(0, \sigma^2/(1 - \alpha^2) + \sigma^2).$$

The claim follows by induction in n .

- (c) We have, for $\ell \in \mathbb{N}_0$,

$$X_\ell = \alpha^\ell X_0 + \sum_{j=1}^{\ell} \alpha^{\ell-j} \sigma Z_j,$$

as we now show via induction: Indeed, the formula is correct for $\ell = 0$, so assume it is correct for $\ell - 1$. Then

$$X_\ell = \alpha X_{\ell-1} + \sigma Z_\ell = \alpha \left(\alpha^{\ell-1} X_0 + \sum_{j=1}^{\ell-1} \alpha^{\ell-1-j} \sigma Z_j \right) + \sigma Z_\ell = \alpha^\ell X_0 + \sum_{j=1}^{\ell} \alpha^{\ell-j} \sigma Z_j,$$

proving the explicit formula for X_ℓ inductively. Therefore,

$$X_n - \alpha^k X_{n-k} = \sum_{j=n-k+1}^n \alpha^{n-j} \sigma Z_j$$

Now we see that

$$\text{Cov}[X_n, \alpha^{-k} X_{n-k}] = \alpha^{-k} \text{Cov} \left[X_{n-k}, X_{n-k} + \sum_{j=n-k+1}^n \alpha^{n-k-j} \sigma Z_j \right] = \text{Var}[X_{n-k}] = \frac{\sigma^2}{1 - \alpha^2}$$

by the bilinearity of the covariance (and using also symmetry), and using in the last step the result of (b). Finally, again by the bilinearity:

$$\text{Cov}[X_n, X_{n-k}] = \alpha^k \text{Cov}[X_n, \alpha^{-k} X_{n-k}] = \frac{\alpha^k \sigma^2}{1 - \alpha^2}.$$

3. Applications of Jensen's inequality

- (a) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be convex and $n \geq 1$. Show that for any $\lambda_1, \dots, \lambda_n \geq 0$ with $\sum_{j=1}^n \lambda_j = 1$, one has

$$g \left(\sum_{j=1}^n \lambda_j x_j \right) \leq \sum_{j=1}^n \lambda_j g(x_j), \quad x_1, \dots, x_n \in \mathbb{R}.$$

(b) Use (a) and the fact that log is concave to establish the *arithmetic mean-geometric mean inequality*:

$$\left(\prod_{j=1}^n x_j \right)^{\frac{1}{n}} \leq \frac{\sum_{j=1}^n x_j}{n}, \quad x_1, \dots, x_n \geq 0.$$

(c) Let $X \geq 0$ be a non-negative real random variable. Assume that all moments of X exist. Show that

$$\mathbf{E}[X] \leq (\mathbf{E}[X^2])^{\frac{1}{2}} \leq (\mathbf{E}[X^3])^{\frac{1}{3}} \leq \dots$$

(a) We consider the random variable X with $\Omega_X = \{x_1, \dots, x_n\}$ and $\mathbf{P}[X = x_i] = \lambda_i$ (if some of the x_j coincide, we write

$$\mathbf{P}[X = x_j] = \sum_{i: x_i = x_j} \lambda_i).$$

Clearly $p_X(x_i) = \mathbf{P}_X[\{x_i\}]$ is a probability mass function. By Jensen's inequality

$$g\left(\sum_{j=1}^n \lambda_j x_j\right) = g(\mathbf{E}[X]) \leq \mathbf{E}[g(X)] = \sum_{j=1}^n g(x_j) \lambda_j.$$

(b) Note that the inequality is trivial if any of the x_j is zero. So we assume $x_j > 0$ for every $1 \leq j \leq n$. By (a) and the fact that log is concave (so $-\log$ is convex), we have (setting $\lambda_j = \frac{1}{n}$)

$$\log\left(\sum_{j=1}^n \frac{1}{n} x_j\right) \geq \sum_{j=1}^n \log(x_j) \frac{1}{n} = \log\left(\left(\prod_{j=1}^n x_j\right)^{\frac{1}{n}}\right).$$

Since exp is strictly increasing, we obtain by applying exp on both sides

$$\frac{1}{n} \sum_{j=1}^n x_j \geq \left(\prod_{j=1}^n x_j\right)^{\frac{1}{n}}.$$

(c) Consider the function $\varphi(x) = x^{\frac{n+1}{n}}$ for $x \geq 0$ and $n \in \mathbb{N}$. This function is convex, since

$$\varphi'(x) = \left(\frac{n+1}{n}\right)x^{\frac{1}{n}} \Rightarrow \varphi''(x) = \frac{n+1}{n} \cdot \frac{1}{n} x^{\frac{1-n}{n}} \geq 0.$$

By applying Jensen's inequality, we have

$$\mathbf{E}\left[Y^{\frac{n+1}{n}}\right] = \mathbf{E}[\varphi(Y)] \geq \varphi(\mathbf{E}[Y]) = \mathbf{E}[Y]^{\frac{n+1}{n}}.$$

We then insert $Y = X^n$ and therefore:

$$\mathbf{E}\left[(X^n)^{\frac{n+1}{n}}\right] \geq \mathbf{E}[X^n]^{\frac{n+1}{n}} \Rightarrow \mathbf{E}[X^{n+1}]^{\frac{1}{n+1}} \geq \mathbf{E}[X^n]^{\frac{1}{n}}.$$

4. (*) Chernoff bounds reloaded

[4 Points]

In this problem you will prove the following theorem:

Let $X_1, \dots, X_n$ independent Bernoulli-distributed random variables with $\mathbf{P}[X_i = 1] = p_i$ and $\mathbf{P}[X_i = 0] = 1 - p_i$. Then, one has for $X = \sum_{i=1}^n X_i$ and $\mu = \mathbf{E}[X] = \sum_{i=1}^n p_i$, and every $\delta > 0$, that $\mathbf{P}[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{1+\delta}}\right)^\mu$.

(a) Prove the previous assertion. Use the following steps:

- Recall problem 3 on Problem set 7, and apply the result to X .
- Replace X by the sum of X_i . Calculate explicitly the expectation appearing in the formula obtained.
- Replace $1 + p_i(e^t - 1)$ with an upper bound (via $1 + y \leq e^y$) and simplify.
- Find t so that the bound is minimal.

- (b) Application: Consider the n -fold toss of a fair coin. Calculate a bound for the probability that the number of “heads” exceeds its expectation by at least 10%, using
- the Čebyšev-inequality,
 - the version of the Chernoff bound found above,
- if $n = 10^3$ or $n = 10^4$. Compare the results.
Hint: Since $p = 1/2$, the distribution is symmetric.

- (a) i. We have

$$\mathbf{P}[X \geq (1 + \delta)\mu] \leq \frac{\mathbf{E}[e^{tX}]}{e^{t(1+\delta)\mu}} = \frac{\psi_X(t)}{e^{t(1+\delta)\mu}},$$

for any $t > 0$, using the result of problem 3 (a) on Problem set 7.

- ii. By independence, we have

$$\psi_X(t) = \mathbf{E}[e^{tX}] = \mathbf{E}\left[e^{t\sum_{i=1}^n X_i}\right] = \prod_{i=1}^n \psi_{X_i}(t).$$

We explicitly calculate:

$$\psi_{X_i}(t) = e^{t \cdot 1} p_i + e^{t \cdot 0} (1 - p_i) = 1 + p_i(e^t - 1).$$

- iii. As suggested, we have

$$1 + p_i(e^t - 1) \leq e^{p_i(e^t - 1)}.$$

Therefore:

$$\mathbf{P}[X \geq (1 + \delta)\mu] \leq \frac{\prod_{i=1}^n (1 + p_i(e^t - 1))}{e^{t(1+\delta)\mu}} \leq \frac{e^{\sum_{i=1}^n p_i(e^t - 1)\mu}}{e^{t(1+\delta)\mu}} = \frac{e^{(e^t - 1)\mu}}{e^{t(1+\delta)\mu}} =: f(t).$$

- iv. We now optimize the function f in $t > 0$. First, we see that $f(0) = 1$, and $\lim_{t \rightarrow \infty} f(t) = \infty$. Moreover

$$f'(t) = f(t)\mu(e^t - (1 + \delta)).$$

We see that the choice $f(t) = \log(1 + \delta)$ yields a local extremum, which due to the shape of the function f and its smoothness must be a local minimum. We insert the value into the formula in iii. and see

$$\mathbf{P}[X \geq (1 + \delta)\mu] \leq \frac{e^{(e^{\log(1+\delta)} - 1)\mu}}{e^{\log(1+\delta)(1+\delta)\mu}} = \frac{e^{\delta\mu}}{(1 + \delta)^{(1+\delta)\mu}},$$

as we claimed.

- (b) We have $\delta = 1/10$, $\mu = n/2$. With Čebyšev:

$$\mathbf{P}[X \geq (1+\delta)\mu] = \mathbf{P}[X - \mu \geq \delta\mu] \stackrel{\text{symm.}}{=} \frac{1}{2} \mathbf{P}[|X - \mu| \geq \mu\delta] \leq \frac{\text{Var}[X]}{2\delta^2\mu^2} = \frac{n/4}{2 \cdot 1/100 \cdot n^2/4} = \frac{50}{n}.$$

With this, we see that we obtain the following bounds/values:

$n =$	10^3	10^4
Čebyšev	0.05	0.005
Chernoff	0.09	$3.1 \cdot 10^{-11}$
exact	0.00087	$7.8 \cdot 10^{-24}$